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ORIGINAL ARTICLE

Effect of permissive hypercarbia on lung oxygenation during one-lung ventilation and postoperative pulmonary complications in patients undergoing thoracic surgery

A prospective randomised controlled trial

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BACKGROUND The effect of hypercarbia on lung oxygenation during thoracic surgery remains unclear.

OBJECTIVE To investigate the effect of hypercarbia on lung oxygenation during one-lung ventilation in patients undergoing thoracic surgery and evaluate the incidence of postoperative pulmonary complications.

DESIGN Prospective randomised controlled trial.

SETTING A tertiary university hospital in the Republic of Korea from November 2019 to December 2020.

PATIENTS Two hundred and ninety-seven patients with American Society of Anaesthesiologists physical status II to III, scheduled to undergo elective lung resection surgery.

INTERVENTION Patients were randomly assigned to Group 40, 50, or 60. An autoflow ventilation mode with a lung protective ventilation strategy was applied to all patients. Respiratory rate was adjusted to maintain a partial pressure of arterial carbon dioxide of 40 ± 5 mmHg in Group 40, 50 ± 5 mmHg in Group 50 and 60 ± 5 mmHg in Group 60 during one-lung ventilation and at the end of surgery.

MAIN OUTCOME MEASURES The primary outcome was the arterial oxygen partial pressure/fractional inspired oxygen ratio after 60 min of one-lung ventilation.

RESULTS Data from 262 patients were analysed. The partial pressure/fractional inspired oxygen ratio was significantly higher in Group 50 and Group 60 than in Group 40 (269.4 vs. 262.9 vs. 214.4; $P < 0.001$) but was not significantly different between Group 50 and Group 60. The incidence of postoperative pulmonary complications was comparable among the three groups.

CONCLUSION Permissive hypercarbia improved lung oxygenation during one-lung ventilation without increasing the risk of postoperative pulmonary complications or the length of hospital stay.

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KEY POINTS

- The effect of hypercarbia on lung oxygenation during one-lung ventilation and postoperative pulmonary complications in patients undergoing thoracic surgery remains unclear.
- The partial pressure/fractional inspired oxygen ratio was significantly lower when a P_aCO_2 of 40 ± 5 mmHg was maintained compared with a P_aCO_2 of 50 ± 5 mmHg or 60 ± 5 mmHg; moreover, the incidence of postoperative pulmonary complications (PPCs) did not differ according to P_aCO_2.
- During one-lung ventilation in thoracic surgery, permissive hypercarbia was associated with improved oxygenation.

Introduction

One-lung ventilation (OLV) is inevitable in thoracic surgery. However, OLV may expose patients to hypoxemia from increased venous shunt and airway dead space,[1,2] and it is a known risk factor for acute lung injury as a result of increase in airway pressure due to collapse of the nondependent lung.[3] Various methods have been used

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to improve oxygenation. These involve the modification of ventilation strategies[4] as well as medical treatments, such as increasing inspiratory oxygen fraction (FiO_2), alveolar recruitment,[5] applying positive end-expiratory pressure (PEEP) to the dependent lung,[6,7] changing the inspiratory-to-expiratory (I:E) ratio,[8] applying continuous positive airway pressure,[9] applying apnoeic oxygen insufflation[10] to the nondependent lung and the administration of nitric oxide,[11] dexmedetomidine[12,13] or iloprost.[14]

Clinically, the normal carbon dioxide level is considered 35 to 45 mmHg, and hypercapnia can be defined as mild (up to 50 mmHg), moderate (55 to 70 mmHg) and severe (higher than 75 mmHg) hypercapnia.[15] Hypercarbia is known to improve peripheral tissue oxygenation in healthy patients,[16,17] and obese surgical patients.[18] Although preliminary studies have been conducted on hypercarbia and its complications, and pilot studies have been conducted on haemodynamic changes during thoracic surgery,[19,20] the relationship between hypercarbia and lung oxygenation during OLV has only been studied in a small number of patients.[21]

The incidence of postoperative pulmonary complications (PPCs) in patients undergoing thoracic surgery is approximately 20%,[3] and the complication rate in patients undergoing lung resection surgery with lung-protective ventilation (LPV) is as high as 13.4%.[22] Nevertheless, efforts have been made to reduce the occurrence of PPCs in patients receiving mechanical ventilation under general anaesthesia,[23,24] and the incidence of PPCs has been reduced using the LPV strategy during OLV.[22,25,26] Therapeutic hypercarbia reduces systemic inflammation during OLV[27]; however, no studies to date have investigated the effect of hypercarbia on PPCs.

Therefore, this study aimed to investigate the effect of hypercarbia on lung oxygenation in patients undergoing thoracic surgery, as well as evaluating the incidence of PPCs related to hypercarbia during thoracic surgery. Our primary hypothesis was that permissive hypercarbia could improve lung oxygenation during OLV.

Methods

Ethics

Ethical approval for this study (No. 4-2019-0904) was provided by the Institutional Review Board (Chairperson Dr Young-Guk Ko) of Yonsei University Health System, Seoul, Republic of Korea, on 5 November 2019. This prospective randomised controlled trial was registered at ClinicalTrials.gov (NCT 04175379). Written informed consent was obtained from all patients before they entered the trial. This manuscript adheres to the applicable Consolidated Standards of Reporting Trials guidelines.

Patient population

Patients who were scheduled to undergo elective lobectomy or segmentectomy with an expected OLV duration longer than 1 h were screened between November 2019 and December 2020. Eligible patients were those aged 40 to 80 years with an ASA physical status of II to III. The exclusion criteria were as follows: severe heart failure defined as New York Heart Association class III or IV, arrhythmia or atrial fibrillation; moderate or severe obstructive lung disease or restrictive lung disease in the preoperative lung function test; low carbon monoxide diffusing capacity less than 75%; history of brain disease or increased intracranial pressure; liver disease with aspartate transaminase levels at least 100 IU ml^{-1} or alanine aminotransferase levels at least 50 IU l^{-1}; renal disease with creatine levels at least 1.5 mg dl^{-1}; preexisting hypercarbia or metabolic acidosis; obesity as defined by a BMI greater than 30 kg m^{-2}; previous history of contralateral lung surgery; pregnancy; inability to communicate effectively.

Anaesthesia management and procedure

Standard monitoring, including pulse oximetry, electrocardiography, and noninvasive blood pressure monitoring was commenced when patients arrived at the operating room. After preoxygenation, anaesthesia was induced using propofol (1.0 to 2.0 mg kg^{-1}), remifentanil (0.5 to 1.0 μg kg^{-1}) and rocuronium (0.6 mg kg^{-1}). All patients were intubated with left-sided double-lumen tubes (DLTs) (VentiBronc™ Anchor; Flexicare Medical Ltd., Mountain Ash, UK).[26] The DLT position was confirmed using a fibreoptic bronchoscope (Karl Storz and Co., Tuttlingen, Germany). An arterial cannula was inserted in the patient's dependent radial artery, and a 7F central vein catheter (Arrow; Teleflex Incorporated, Wayne, Pennsylvania, USA) was inserted in the right internal jugular vein.

Anaesthesia was maintained using 1.0 to 2.0 vol% sevoflurane and 0.05 to 0.1 μg kg^{-1} min^{-1} remifentanil to maintain a target SedLine (Masimo Corp., Irvine, California, USA) value between 25 and 50. Autoflow ventilation mode (Primus; Dräger, Lübeck, Germany) was used in all patients during anaesthesia. Ventilation was initiated using a tidal volume of 6 ml kg^{-1} of the predicted body weight with an I : E ratio of 1 : 2, FiO_2 of 0.6, and PEEP of 5 cmH_2O. Baseline arterial blood gas (ABG) analysis was performed in the supine position while maintaining an end-tidal CO_2 ($EtCO_2$) of 40 ± 5 mmHg during two-lung ventilation (TLV).

Hypotension was defined as a mean blood pressure (MBP) of less than 65 mmHg or less than 20% of the baseline value, and it was treated by administering ephedrine or norepinephrine as required. Ephedrine was administered primarily for patients with low MBP, and norepinephrine was administered when continuous use of vasoactive drugs was required.

Bradycardia was defined as a heart rate (HR) of less than 50 beats per minute, and it was treated by administering atropine (0.25 to 0.5 mg).

Intervention

The patients were randomly assigned to Group 40, 50, or 60 according to a computer-generated randomisation list of 1 : 1 : 1, and their respiratory rates were adjusted to maintain a partial pressure of arterial carbon dioxide (P_aCO_2) 40 ± 5 mmHg, 50 ± 5 mmHg or 60 ± 5 mmHg, respectively, during OLV and until the end of the surgery. After 30 min of OLV in the lateral decubitus position, ABG analysis was performed to determine the difference between the P_aCO_2 and $EtCO_2$. Using this difference, $EtCO_2$ was adjusted to match the target P_aCO_2 by adjusting respiratory rates in group. After 60 min of OLV, ABG analysis was repeated to determine the primary outcome. When the patient's oxygen saturation (SaO_2), measured using pulse oximetry, decreased to less than 90% during OLV, FiO_2 was increased stepwise to 0.8 and 1.0. If the SaO_2 decreased below 90%, even at an FiO_2 of 1.0, TLV was applied and the study was discontinued.

Outcomes

The primary outcome was the arterial oxygen/inspired oxygen ratio (PaO_2/FiO_2 ratio) after 1 h of OLV. The secondary outcome was the incidence of PPCs, defined as the presence of one or more of the following postoperative conditions: peri-operative mortality, tracheostomy, reintubation, initial ventilatory support for greater than 48 h, acute respiratory distress syndrome (ARDS), bronchopleural fistula (BPF), pulmonary embolism, pneumonia, reoperation and myocardial infarction.[28] Peri-operative mortality was defined as death during the same hospitalisation period as the surgery or within 30 days after the surgery. Pneumonia and ARDS were defined as described in a previous study.[22] The length of hospital stay was also evaluated.

The respiratory variables included tidal volume, respiratory rates, minute ventilation, PEEP, FiO_2, $EtCO_2$, peak inspiratory pressure (PIP) and dynamic compliance. These were measured after 1 h of OLV. Dynamic compliance was calculated as TV/(PIP − PEEP). Haemodynamic variables obtained were the MBP, HR and central venous pressure (CVP). Anaesthetic depth was determined using the patient status index (SedLine, Masimo Corp.). P_aCO_2, pH, base excess, and the P_aO_2/FiO_2 ratio were obtained using ABG analysis.

Sample size and statistical analysis

The required sample size was estimated on the basis of the findings of a previous study,[7] wherein a change in the P_aO_2/FiO_2 ratio of 30 mmHg was assumed to be significant. The sample size calculations revealed that to obtain 80% power at the 5% significance level and to allow for a maximum drop-out rate of 10%, 93 patients would have to be randomly assigned to each group (total, 279 patients).

Normality of the data sets was assessed using the Kolmogorov–Smirnov test. All data are expressed as means ± SD, ranges, numbers and percentages, or medians [IQR], as indicated. Data from the three groups were compared using analysis of variance or the Kruskal–Wallis test as appropriate, and post hoc analysis was performed using the Bonferroni correction. All statistical analyses were performed using IBM SPSS Statistics for Windows/Macintosh, Version 26.0 (IBM Corp., Armonk, New York, USA), and P less than 0.05 was considered statistically significant.

Results

Patient characteristics and intraoperative data

Among the 297 patients screened, 279 were randomly allocated to one of the three groups. Thereafter, 17 were excluded for the following reasons: OLV time less than 1 h ($n = 12$), surgical plan changed to wedge resection after confirming the frozen biopsy findings ($n = 4$), and loss to follow-up ($n = 1$). Finally, the data from 262 patients, including 88 in Group 40, 88 in Group 50 and 86 in Group 60, were analysed (Fig. 1). The baseline characteristics, intraoperative data, and ABG analysis results during TLV did not show any significant difference among the three groups (Table 1).

Analyses of the primary and secondary outcomes

Table 2 shows the comparison of intraoperative data during OLV among the three groups. Among the haemodynamic variables, HR was significantly higher in the hypercarbia group than in Group 40 ($P = 0.004$); however, there was no significant difference between Group 50 and Group 60. The MBP and CVP were similar among the three groups. Anaesthetic depth also did not show any significant difference. Among the respiratory parameters, respiratory rate and minute ventilation were the highest in Group 40 and the lowest in Group 60. The PIP was significantly lower and dynamic compliance was significantly higher in Group 50 and Group 60 than in Group 40 ($P < 0.001$). Tidal volume, PEEP and FiO_2 showed no differences among the three groups.

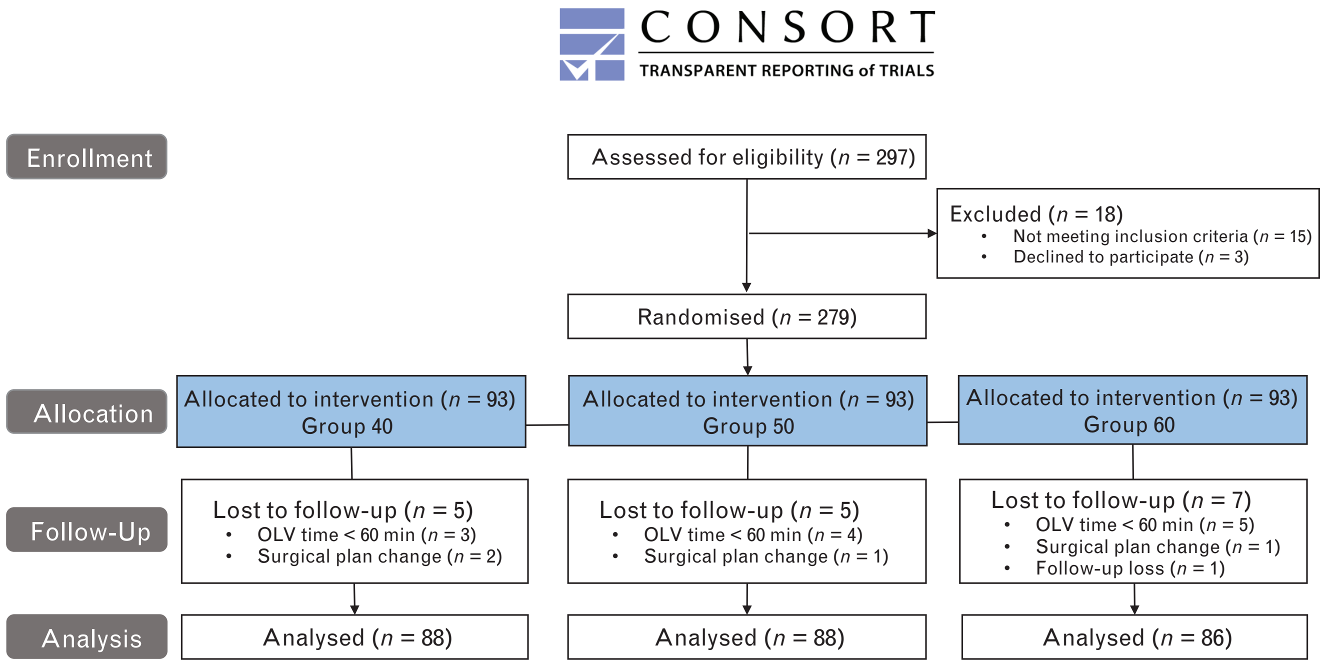
Primary outcome

The P_aO_2/FiO_2 ratio was significantly higher in Group 50 and Group 60 than in Group 40 ($P < 0.001$) but showed no significant difference between Group 50 and Group 60 (Fig. 2). In the ABG analysis, P_aCO_2 and base excess were the highest in Group 60 and the lowest in Group 40, whereas pH was the lowest in Group 60 and the highest in Group 40.

Secondary outcomes

The overall incidence of PPCs was not significantly different among the three groups (Table 3). The incidence of PPCs such as pneumonia, BPF and pulmonary embolism did not show any significant difference among the three groups. PPCs such as initial ventilator care over 48 h, ARDS, reintubation, tracheostomy and myocardial infarction were not observed. However, one patient in

Fig. 1 Consolidated Standards of Reporting Trials flow diagram.



Group 40: P_aCO_2 maintained at 40 ± 5 mmHg during OLV and to the end of surgery. Group 50: P_aCO_2 maintained at 50 ± 5 mmHg during OLV and to the end of surgery. Group 60: P_aCO_2 maintained at 60 ± 5 mmHg during OLV and to the end of surgery. OLV, one-lung ventilation; P_aCO_2, partial pressure of arterial carbon dioxide.

each of the three groups required reoperation for the following reasons: bleeding control in Group 40, empyema in Group 50 and air-leakage control in Group 60. Nevertheless, no mortality occurred during the perioperative period. The length of hospital stay was not significantly different among the three groups.

Discussion

In this prospective randomised controlled study, permissive hypercarbia in a P_aCO_2 range of 50 to 60 mmHg under the LPV strategy improved lung oxygenation during OLV in patients undergoing thoracic surgery. In the hypercarbia group, lung oxygenation improved during OLV, while the incidence of PPCs and the length of hospital stay did not increase. Based on these findings, we recommend the use of mild permissive hypercarbia targeting a P_aCO_2 of 50 to 60 mmHg in patients undergoing thoracic surgery.

A previous sheep model study showed that hypercarbia improved tissue oxygenation and decreased lung oedema.[29] In a human study, mild hypercarbia improved peripheral oxygenation in healthy patients.[16] Moreover, in the surgical setting, LPV reduced PPCs in patients undergoing thoracic surgery[25,26]; however, the relationships between hypercarbia induced by LPV and lung oxygenation have not been extensively investigated. Therefore, in our study, we aimed to elucidate the relationship between hypercarbia and lung oxygenation during OLV in patients undergoing thoracic surgery.

The improvement in oxygenation induced by hypercarbia in this study could be explained as follows. First, hypercarbia may have affected hypoxic pulmonary vasoconstriction (HPV), which is caused by hypoxemia of the nondependent lung during OLV. HPV also improves ventilation/perfusion mismatch by increasing pulmonary blood flow to the dependent lung by causing the contraction of vascular smooth muscles. Animal experiments have shown that the effects of hypercarbia on the pulmonary vasculature are independent of HPV, and that they are known to cause pulmonary vasodilation[30]; however, in contrast, human studies have shown that hypercarbia causes pulmonary vasoconstriction.[31] Moreover, it remains unknown whether these actions occur globally or regionally.[32] Our results suggest that the pulmonary vasoconstriction induced by hypercarbia can improve oxygenation by enhancing HPV in the nondependent lung. However, to obtain a definitive explanation, a lung perfusion scan would need to be performed during OLV, and this would be challenging to execute during surgery. Second, the improvement in oxygenation may be caused by the haemodynamic changes induced by hypercarbia. In this study, HR was significantly increased in the mild hypercarbia group, and this was consistent with the results of previous animal and human studies. In one animal study,[33] hypercarbia increased the cardiac output and HR because of the activation of the sympathetic nervous system. A human study also showed an increase in HR, stroke volume, cardiac output and MBP as a result

Table 1 Baseline characteristics and perioperative data of the three study groups

	Group 40 ($n = 88$)	Group 50 ($n = 88$)	Group 60 ($n = 86$)	P value
Sex (M/F)	49 (56)/39 (44)	39 (44)/49 (56)	40 (47)/46 (53)	0.279
Age (years)	63.4 ± 8.8	63.1 ± 9.1	62.6 ± 9.6	0.843
Height (cm)	162.9 ± 8.1	162.2 ± 7.4	161.6 ± 8.4	0.542
Weight (kg)	64.5 ± 10.3	64.0 ± 9.4	63.2 ± 10.4	0.700
BMI (kg m^{-2})	24.2 ± 2.6	24.2 ± 2.4	24.1 ± 2.9	0.958
Medical history				
HTN	40 (45.5)	32 (36.4)	28 (32.6)	0.197
DM	14 (15.9)	15 (17.0)	17 (19.8)	0.790
Old TB	6 (6.8)	5 (5.7)	7 (8.1)	0.814
Smoking				0.267
Ex-smoker	22 (25.0)	27 (31.0)	18 (20.9)	
Current smoker	11 (12.5)	4 (4.6)	9 (10.5)	
ASA PS				0.749
II	48 (54.5)	43 (48.9)	45 (52.3)	
III	40 (45.5)	45 (51.1)	41 (47.7)	
Pulmonary function test				
FVC (% predicted)	94.9 ± 12.5	96.2 ± 12.6	92.9 ± 11.9	0.201
FEV_1 (% predicted)	96.7 ± 14.2	98.1 ± 15.0	93.4 ± 14.1	0.081
FEV_1/FVC ratio (%)	74 [71.5 to 79]	76 [71 to 80]	75.5 [71 to 80]	0.578
DLCO (% predicted)	97.5 ± 16.7	101.3 ± 16.7	95.2 ± 14.3	0.138
Intraoperative data				
Duration of anaesthesia (min)	153.6 ± 30.5	153.4 ± 38.1	149.9 ± 42.5	0.758
Duration of surgery (min)	103.3 ± 27.1	103.3 ± 32.4	101.1 ± 35.0	0.868
Duration of OLV (min)	91.3 ± 26.5	86.1 ± 25.3	89.3 ± 32.1	0.466
Operation type				
VATS/open thoracotomy	85 (96.6)/3 (3.4)	84 (95.5)/4 (4.5)	83 (96.5)/3 (3.5)	0.908
Lobectomy/segmentectomy	81 (92.0)/7 (8.0)	80 (90.9)/8 (9.1)	73 (84.9)/13 (15.1)	0.260
Operation lobe site				
Right upper/middle/lower	24 (27)/14 (16)/20 (23)	22 (25)/8 (9)/12 (14)	27 (31)/4 (5)/21 (24)	0.180
Left upper/lower	17 (19)/13 (15)	23 (26)/23 (26)	22 (26)/12 (14)	0.423
Fluid intake (ml)	996.0 ± 41.0	984.6 ± 40.2	918.6 ± 37.3	0.332
Urine output (ml)	182.4 ± 15.9	178.9 ± 15.1	174.4 ± 18.1	0.374
Estimated blood loss (ml)	48.4 ± 9.6	69.2 ± 12.7	42.9 ± 9.5	0.187
Use of vasoactive drugs	32 (36.4)	37 (42.0)	31 (36.0)	0.655
ABG during TLV in the supine position				
$PaCO_2$ (mmHg)	40.3 ± 3.3	41.2 ± 3.2	41.3 ± 3.4	0.128
pH	7.38 ± 0.03	7.37 ± 0.03	7.37 ± 0.03	0.07
Base excess (mmol l^{-1})	-1.2 ± 1.7	-1.0 ± 1.7	-0.8 ± 1.9	0.314
PF ratio (kPa)	406.6 ± 72.6	410.9 ± 78.4	411.6 ± 82.1	0.900

Data are presented as the mean ± SD, median [IQR], or n (%) as appropriate. Group 40: P_aCO_2 maintained at 40 ± 5 mmHg during OLV and the end of surgery. Group 50: P_aCO_2 maintained at 50 ± 5 mmHg during OLV and the end of surgery. Group 60: P_aCO_2 maintained at 60 ± 5 mmHg during OLV and the end of surgery. ABG, arterial blood gas analysis; ASA PS, ASA physical status; DLCO, diffusion capacity of lung for carbon monoxide; DM, diabetes mellitus; FEV_1, forced expiratory volume in 1 s; FVC, forced vital capacity; HTN, hypertension; OLV, one-lung ventilation; P_aCO_2, partial pressure of arterial carbon dioxide; PF ratio, PaO_2/FiO_2 ratio; TB, tuberculosis; TLV, two-lung ventilation; VATS, video-assisted thoracic surgery.

of hypercarbia.[16,31] Consistent with another previous study,[31] cardiac output in our study increased as the HR increased in the hypercarbia group, and these haemodynamic changes may have improved oxygenation. Alternatively, the decreased airway pressure in the hypercarbia group possibly resulted in a decrease in intrathoracic pressure, resulting in an increase in venous return and cardiac output. Pulmonary artery catheterisation or transoesophageal echocardiography could be performed to obtain more precise explanations; however, pulmonary artery catheterisation is a highly invasive procedure. Hence, it was not performed in our study.

The overall incidence of PPCs in all three groups in this study was less than 6%, which was lower than that reported in previous studies.[3,22] This could be explained by the findings of a previous animal study, which revealed that hypercarbia could protect the lungs by attenuating pulmonary inflammation via the prevention of ischaemic–reperfusion injury and subsequent free radical production.[34] A human study also reported that hypercarbia reduced systemic inflammation.[27] In addition, although a significant difference in pH was observed between Group 50 and Group 60, the incidence of PPCs was low, and no differences in the PPC incidence was observed among the groups. This could be attributed to the fact that the current study included relatively young and healthy patients, minimally invasive surgery was utilised, and there was a relatively short operation time of approximately 2 h. Moreover, although the PIP was the highest and dynamic compliance was the lowest in Group 40, the incidence of PPCs did not differ among the three groups because the included patients had normal lung function and LPV was used.

This is the first study to elucidate the relationship between hypercarbia and lung oxygenation during OLV. The results suggest that an appropriate target P_aCO_2

Table 2 Comparison of intraoperative data and outcomes among the three groups

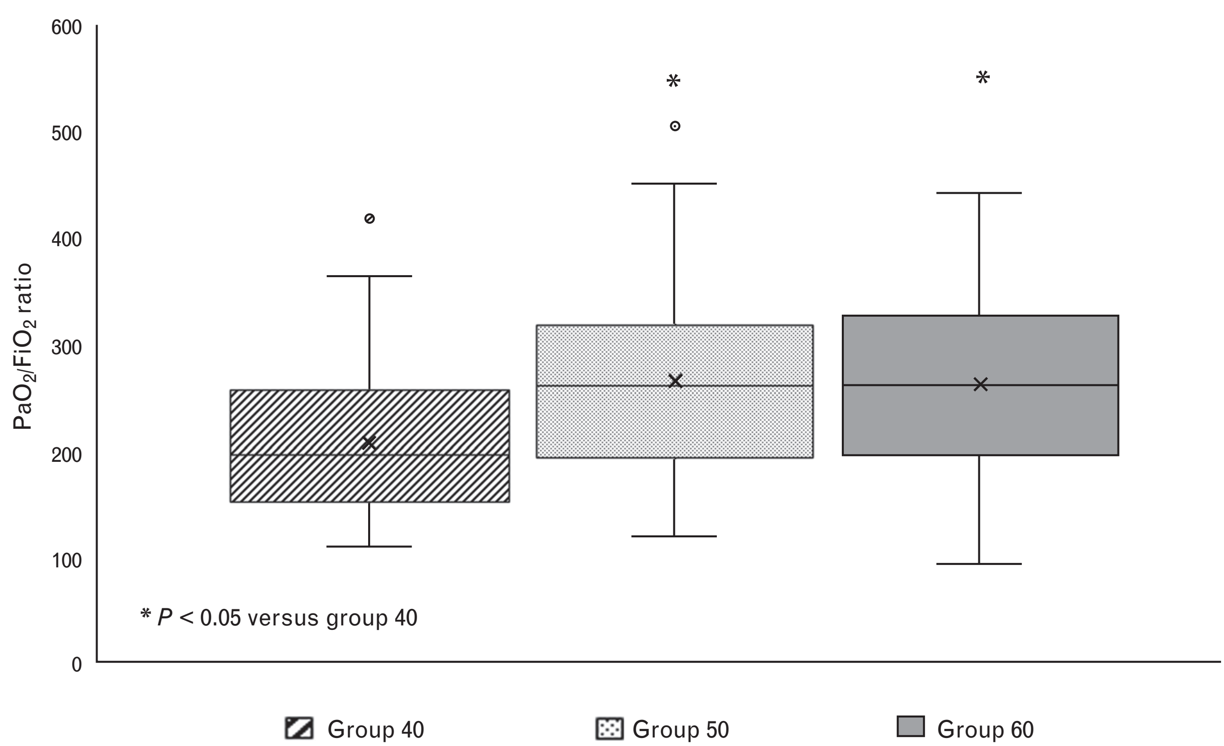
	Group 40 ($n=88$)	Group 50 ($n=88$)	Group 60 ($n=86$)	*P* value
Haemodynamics				
MBP (mmHg)	82.5 ± 9.9	79.7 ± 10.0	79.8 ± 11.3	0.130
HR (min^{-1})	77.4 ± 11.4	82.2 ± 10.0	82.5 ± 12.0	$0.004^{*,\dagger}$
CVP (mmHg)	13.6 ± 3.3	13.3 ± 3.0	14.2 ± 3.4	0.515
Anaesthesia depth				
PSI	38.6 ± 6.7	38.9 ± 7.4	40.5 ± 7.5	0.174
Lung mechanics during OLV				
Tidal volume (ml)	345.8 ± 53.4	339.0 ± 50.3	336.4 ± 55.0	0.480
Respiratory rate (min^{-1})	14.2 ± 2.6	10.2 ± 2.1	7.9 ± 1.8	$<0.001^{*,\dagger,\ddagger}$
Minute ventilation ($l\,min^{-1}$)	4.9 ± 1.0	3.4 ± 0.7	2.7 ± 0.7	$<0.001^{*,\dagger,\ddagger}$
PEEP (cmH_2O)	5	5	5	>.999
FiO_2	0.70 ± 0.14	0.66 ± 0.13	0.69 ± 0.14	0.160
PIP (cmH_2O)	19 [17 to 21]	17 [15 to 18]	16 [15 to 17]	$<0.001^{*,\dagger}$
Dynamic compliance ($ml\,cmH_2O^{-1}$)	18.4 [15.8 to 20.0]	20.0 [17.7 to 23.6]	21.3 [18.7 to 24.2]	$<0.001^{*,\dagger}$
ABG during OLV				
$PaCO_2$ (mmHg)	40.8 ± 3.7	51.7 ± 3.1	60.9 ± 4.0	$<0.001^{*,\dagger,\ddagger}$
pH	7.37 ± 0.03	7.31 ± 0.03	7.26 ± 0.02	$<0.001^{*,\dagger,\ddagger}$
Base excess ($mmol\,l^{-1}$)	-1.4 ± 1.6	-0.4 ± 1.8	0.1 ± 2.1	$<0.001^{*,\dagger,\ddagger}$
PF ratio	214.4 ± 69.3	269.4 ± 84.2	262.9 ± 80.9	$<0.001^{*,\dagger}$

Data are presented as the mean ± SD or median [IQR], or *n* (%) as appropriate. Group 40: P_aCO_2 maintained at 40 ± 5 mmHg during OLV and the end of surgery. Group 50: P_aCO_2 maintained at 50 ± 5 mmHg during OLV and the end of surgery. Group 60: P_aCO_2 maintained at 60 ± 5 mmHg during OLV and the end of surgery. ABG, arterial blood gas analysis; CVP, central venous pressure; FiO_2, inspired oxygen fraction; HR, heart rate; MBP, mean blood pressure; OLV, one-lung ventilation; $PaCO_2$, partial pressure of arterial carbon dioxide; PEEP, positive end-expiratory pressure; PIP, peak inspiratory pressure; PSI, patient status index; PF ratio, P_aO_2/FiO_2 ratio. $^*P<0.05$ Group 40 versus Group 50. $^\dagger P<0.05$ Group 40 vs. Group 60. $^\ddagger P<0.05$ Group 50 vs. Group 60.

level of 50 to 60 mmHg is a more effective OLV strategy and is not associated with more pulmonary complications.

The current study has some limitations. First, we used autoflow mode in patients with normal lung function. Applying autoflow mode during OLV is recommended in terms of lowering PIP and reducing barotrauma,[35] but the PIP obtained by reducing respiratory rate in autoflow mode may not accurately reflect the alveolar pressure. On the other hand, static compliance obtained from volume

Fig. 2 Effects of hypercarbia on the P_aO_2/FiO_2 ratio during one-lung ventilation.



PF ratio, PaO_2/FiO_2 ratio, the ratio of arterial oxygen partial pressure to inspired oxygen fraction. $^*P<0.05$ vs. Group 40.

Table 3 **Comparison of the incidence of postoperative pulmonary complications among the three groups**

	Group 40 ($n = 88$)	Group 50 ($n = 88$)	Group 60 ($n = 86$)	*P* value
PPCs	5 (5.7)	4 (4.5)	3 (3.5)	0.787
Initial ventilator care >48 h	0	0	0	N/A
Pneumonia	3 (3.4)	3 (3.4)	2 (2.3)	0.892
ARDS	0	0	0	N/A
BPF	0	1 (1.1)	0	0.371
Pulmonary embolism	1 (1.1)	0	0	0.371
Reintubation	0	0	0	N/A
Tracheostomy	0	0	0	N/A
Myocardial infarction	0	0	0	N/A
Reoperation	1 (1.1)	1 (1.1)	1 (1.2)	>.999
Bleeding	1 (1.1)	0	0	
Empyema	0	1 (1.1)	0	
Air-leakage control	0	0	1 (1.2)	
Mortality [*n* (%)]	0	0	0	N/A
Length of hospital stay (days)	7.0 [5.0 to 8.8]	6.0 [5.0 to 8.0]	6.0 [5.0 to 8.0]	0.539

Data are presented as the median [IQR] or *n* (%) as appropriate. Group 40: P_aCO_2 maintained at 40 ± 5 mmHg during OLV and the end of surgery. Group 50: P_aCO_2 maintained at 50 ± 5 mmHg during OLV and the end of surgery. Group 60: P_aCO_2 maintained at 60 ± 5 mmHg during OLV and the end of surgery. ARDS, acute respiratory distress syndrome; BPF, bronchopleural fistula; N/A, not available; PPCs, postoperative pulmonary complications.

controlled ventilation mode reflects the distending pressure of the alveoli and is considered a better indicator for lung injury than peak airway pressure. Hence, in future studies in patients with poor lung conditions, volume controlled ventilation would be a better way to measure static compliance. Second, as this study included only healthy patients, further studies are warranted to determine the effects of hypercarbia in patients with systemic disease, which can be affected by respiratory acidosis like pulmonary disease, cardiovascular disease or central nervous disorder. Third, evaluation of the time-dependent effects of hypercarbia-induced acidosis was difficult because the OLV time and surgery time were relatively short. Fourth, the surgical effect on P_aO_2 or P_aCO_2 was not considered. Surgical variation would have been trivial in the study, and hence there was no difference in the type of surgery, direction of resection and operative time among the three groups.

Conclusion

This study demonstrated that permissive hypercarbia improved lung oxygenation during OLV without increasing the risk of PPCs or the length of hospital stay. These results should inspire further research on the clinical effects of hypercarbia in patients with impaired lung disease undergoing thoracic surgery.

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